Project Title:

Sim2Terrain AI : Synthetic Data Driven Off-Road Terrain Segmentation for Autonomous Navigation

Team Name : IgniteX

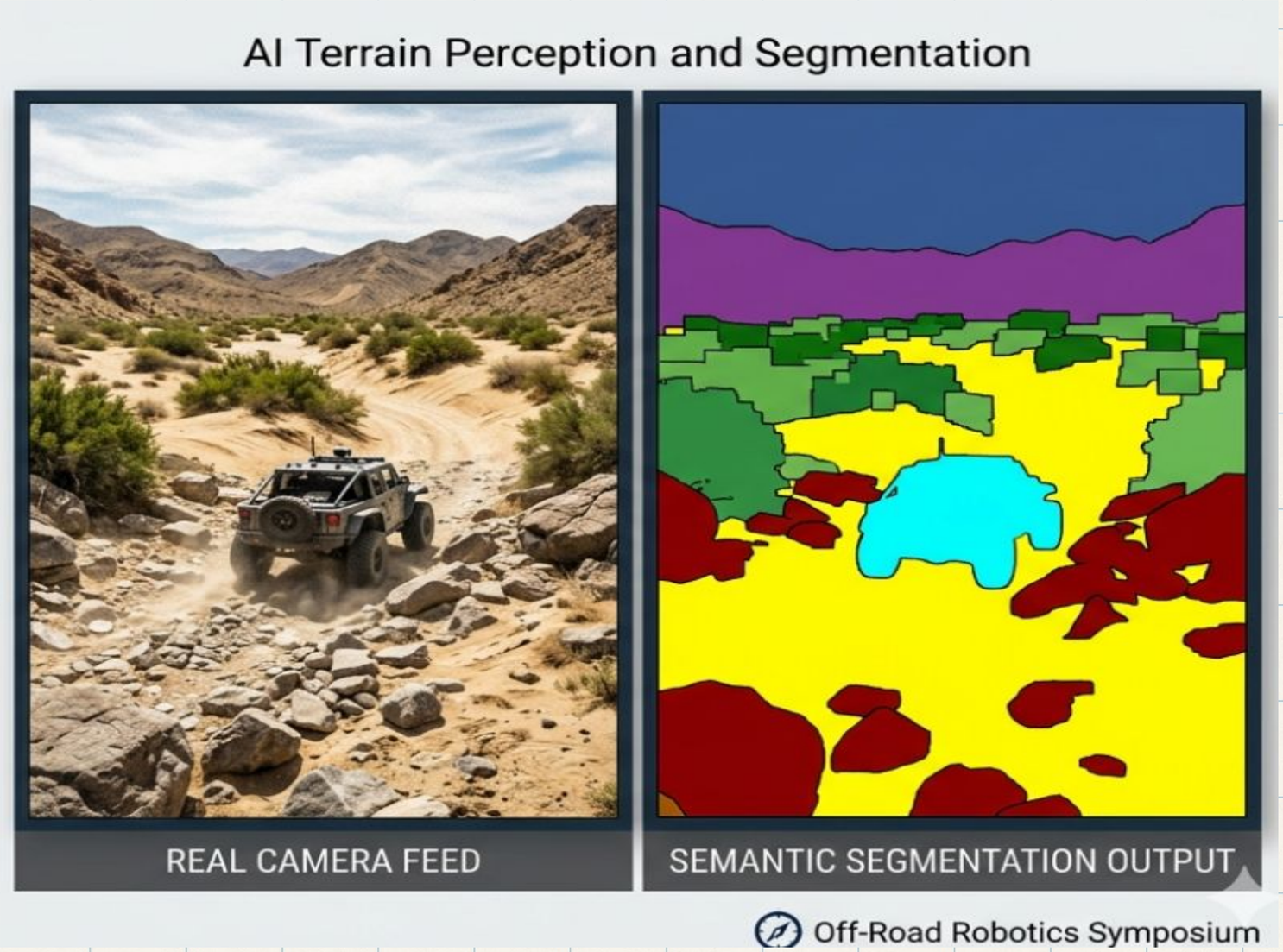
Track : 1 (2nd year)

Problem Statement

UNSTRUCTURED ENVIRONMENTS:
Complex terrain with varying rock sizes, loose sand, and scrub vegetation.

PERCEPTION DIFFICULTY:
Lack of lane markings or clear paths complicates obstacle detection and path planning.

DATA COLLECTION COSTS:
Gathering large-scale, annotated datasets in remote desert and rugged areas is expensive and time-consuming.



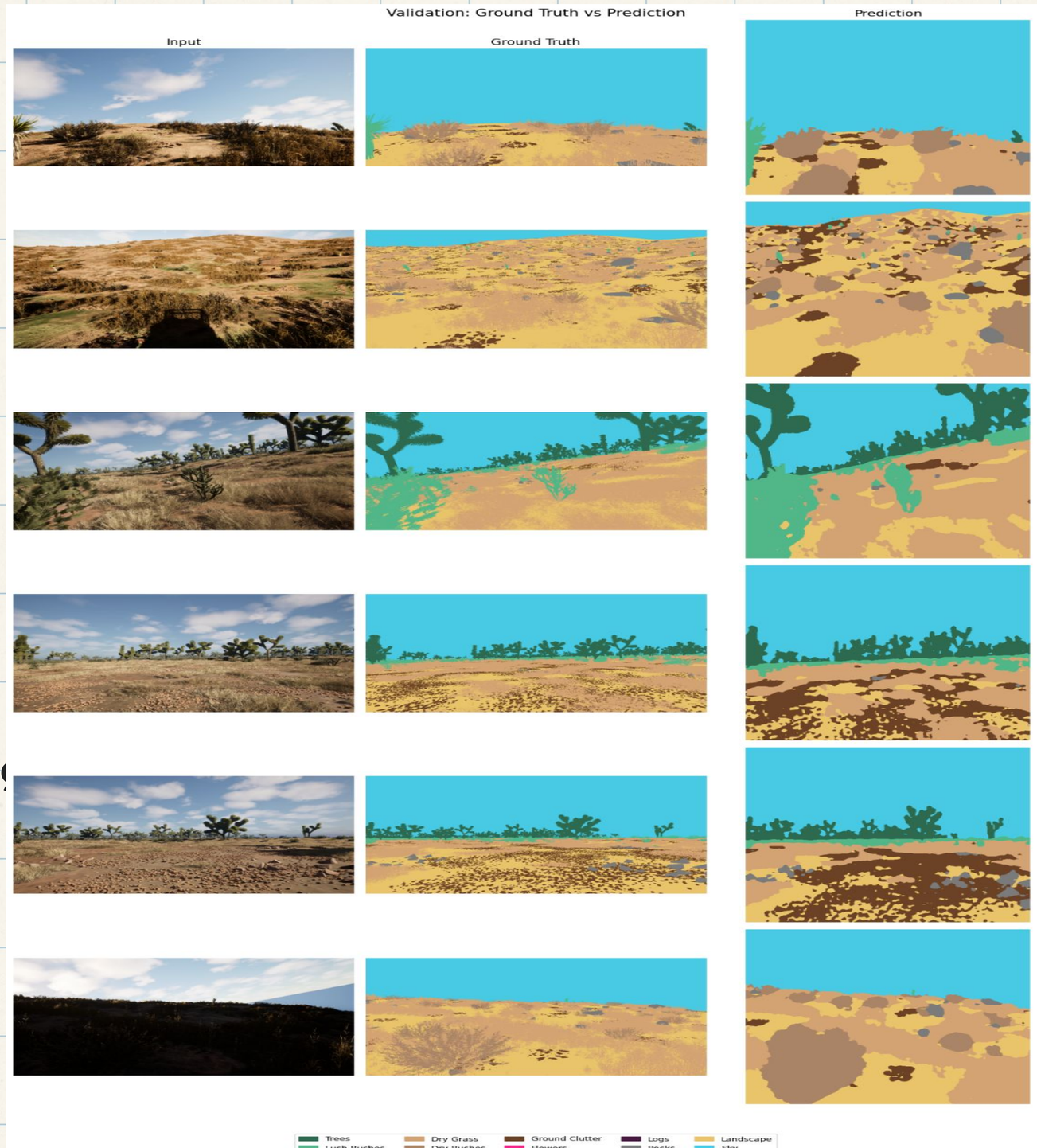
Solution

Generate Synthetic Data: Use the Falcon digital twin simulation platform to create labeled desert terrain images.

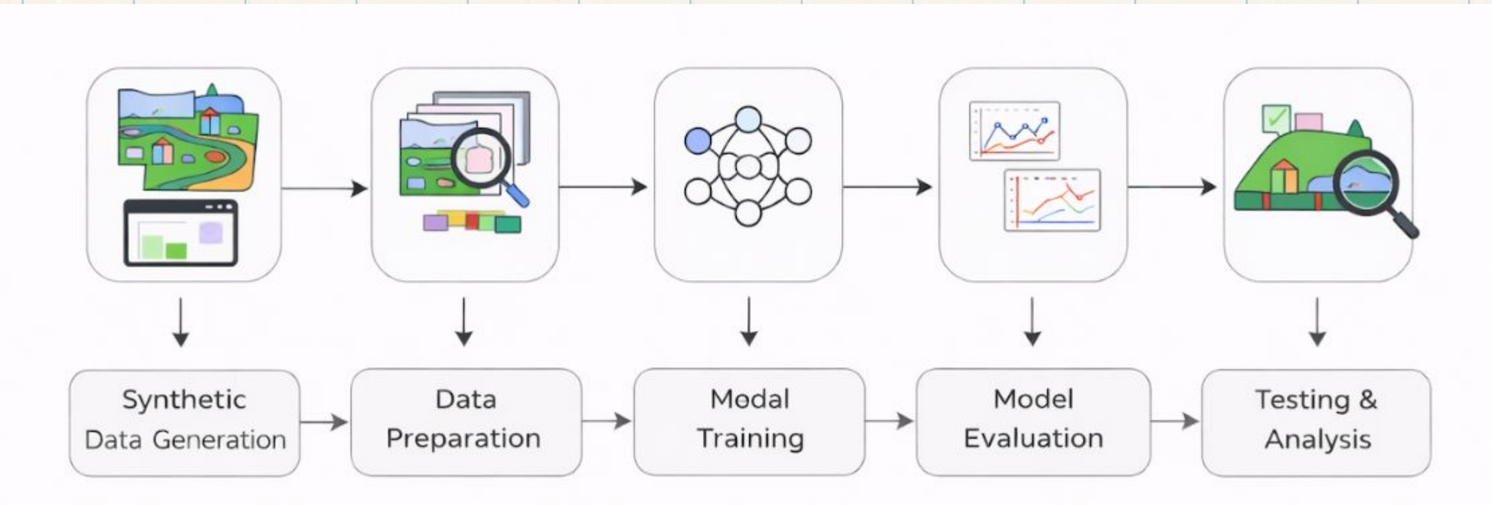
Train AI Model: Train a semantic segmentation model to classify terrain types such as sand, rocks, vegetation, and obstacles.

Model Evaluation: Evaluate performance using IoU score and loss graphs.

Testing & Analysis: Test the model on new unseen environments and analyze results to improve accuracy



Methodology



Synthetic Data Generation
Generate labeled terrain images using Falcon simulation.

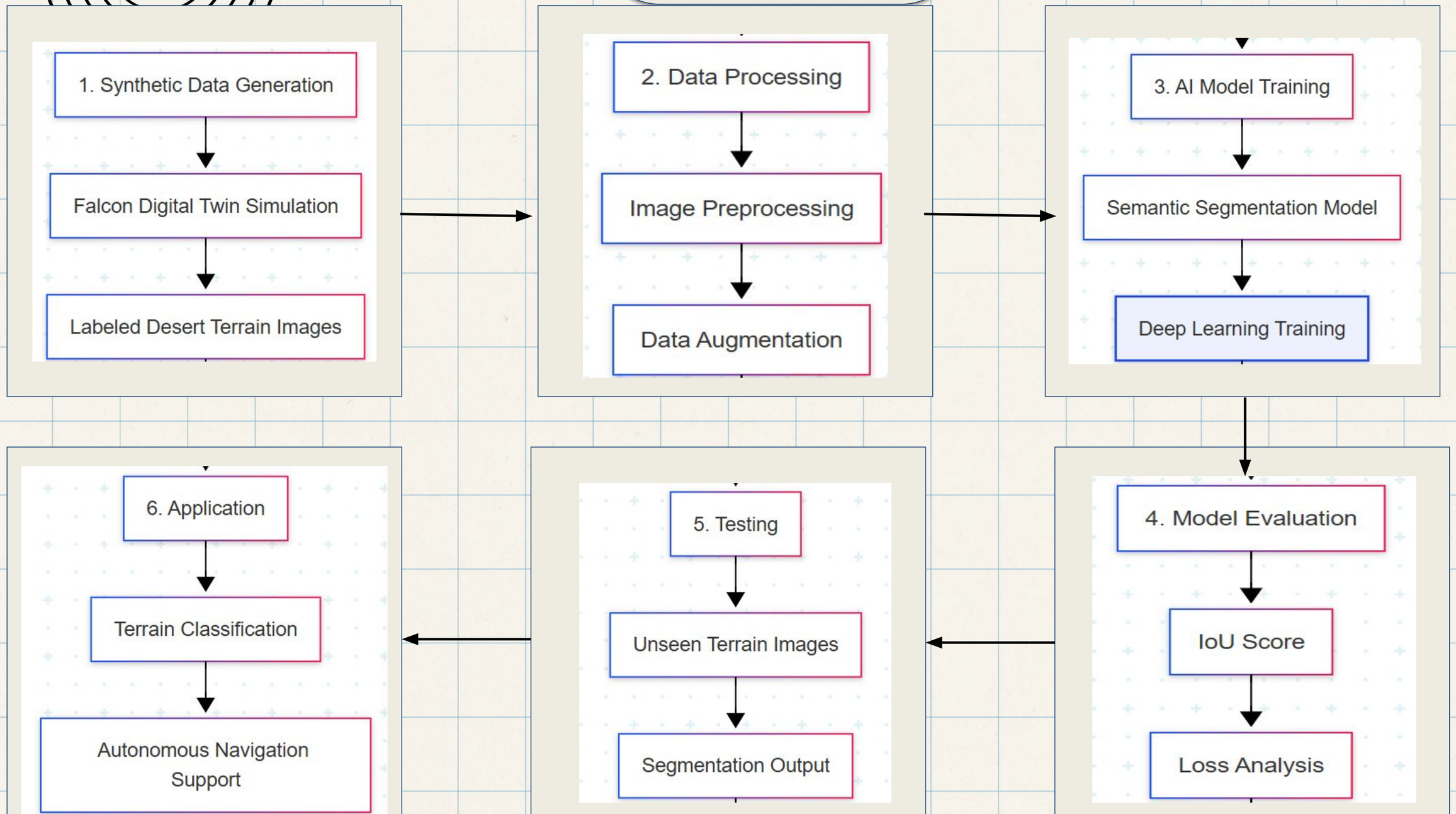
Data Preparation
Apply data augmentation and filtering to improve dataset quality.

Model Training
Train a deep learning semantic segmentation model

Model Evaluation
Measure performance using IoU score and loss graphs

Testing & Analysis
Test the model on unseen environments and analyze results.

Workflow


1. Synthetic Data Generation
Falcon Digital Twin Simulation
Labeled Desert Terrain Images
2. Data Processing
Image Preprocessing
Data Augmentation
3. AI Model Training
Semantic Segmentation Model
Deep Learning Training
4. Model Evaluation
IoU Score
Loss Analysis
5. Testing
Unseen Terrain Images
Segmentation Output
6. Application
Terrain Classification
Autonomous Navigation Support

Technologies Used

Technology Used :

Falcon Digital Twin Simulation Platform : for generating synthetic desert terrain data
Python :for model development and training
PyTorch : deep learning framework for training the model
OpenCV :for image preprocessing
NumPy & Pandas : for data handling and processing

Model Used :

DeepLabV3 with ResNet-101 Backbone for semantic segmentation of terrain images

Evaluation Metrices:

Intersection over Union (IoU)
Loss Graph Analysis

Result & Accuracy

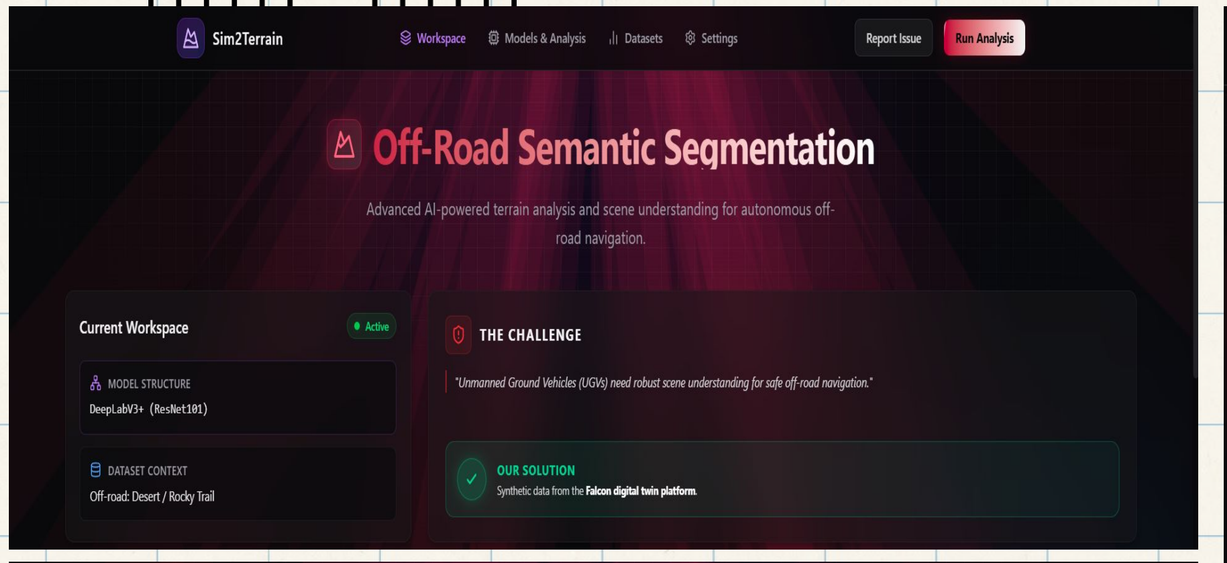
- Trained a semantic segmentation model using Falcon synthetic dataset
- Model successfully segmented terrain classes in desert environments
- Performance evaluated using IoU score and loss metrics
- Model tested on novel unseen desert environments to check generalization

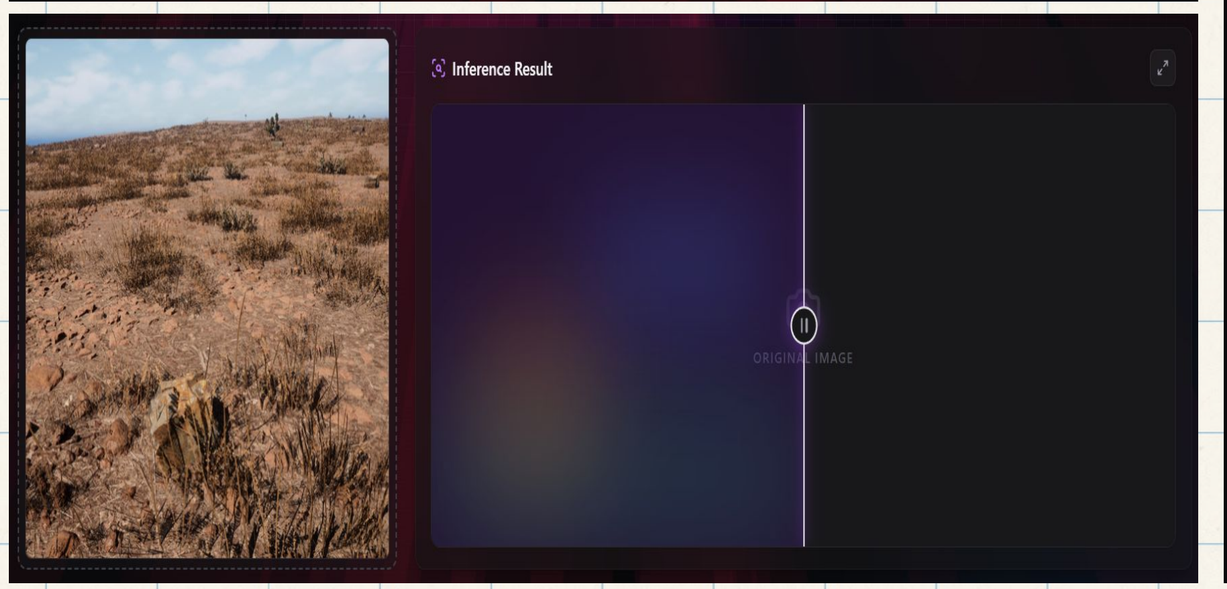
```
...  Best model: epoch 30,  val mIoU = 0.6012

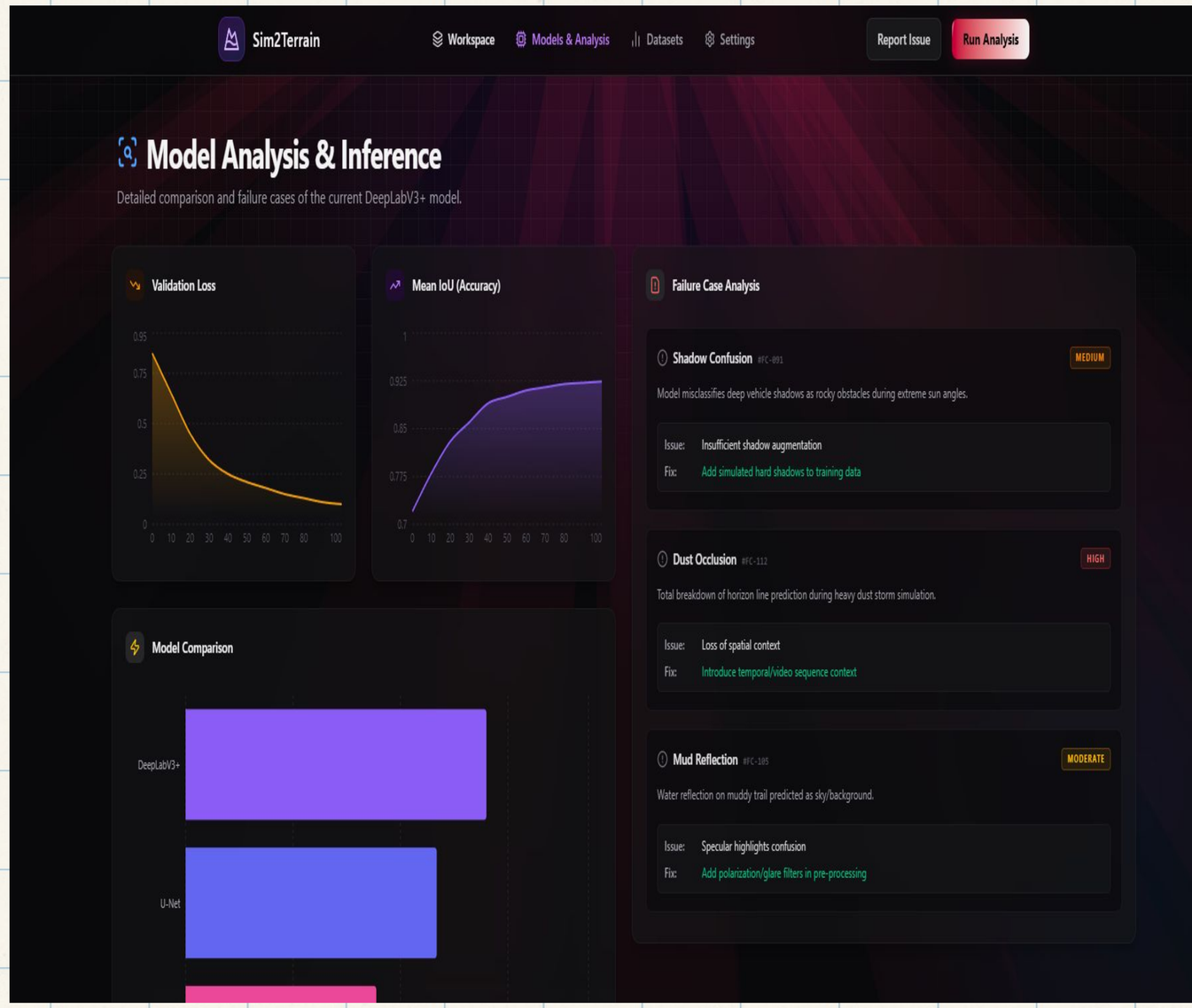
📊 Final Validation Results
  Pixel Accuracy  : 0.8327  (83.27%)
  Mean IoU (mIoU) : 0.6012

  Class                IoU
  ----------------------------
  Trees              0.8180
  Lush Bushes        0.6395
  Dry Grass          0.6547
  Dry Bushes         0.4431
  Ground Clutter     0.3250
  Flowers            0.5954
  Logs               0.5243
  Rocks              0.4512
  Landscape          0.5818
  Sky                0.9793
```

Result & Accuracy







Challenges

Challenges

- Lack of large real-world labeled off-road datasets
- Domain shift between training and new environments
- Achieving high segmentation accuracy

Solutions

- Used Falcon digital twin simulation to generate synthetic data
- Tuned training parameters and model optimization
- Evaluated model using IoU and loss analysis

Before

```
Starting evaluation on test data...
Evaluation complete. Calculating metrics...

--- Test Metrics ---
Pixel Accuracy: 0.5582
Mean Precision: 0.4325
Mean Recall:    0.3418
Mean F1 Score:  0.3421
mIoU:           0.2630
--------------------
Per-class metrics:
Class 0: Precision=0.7385, Recall=0.2725, F1=0.3981, IoU=0.2485
Class 1: Precision=0.0000, Recall=0.0021, F1=0.0000, IoU=0.0000
Class 2: Precision=0.7034, Recall=0.2514, F1=0.3704, IoU=0.2273
Class 3: Precision=0.0000, Recall=0.0000, F1=0.0000, IoU=0.0000
Class 4: Precision=0.0000, Recall=0.0000, F1=0.0000, IoU=0.0000
Class 5: Precision=0.0000, Recall=0.0000, F1=0.0000, IoU=0.0000
Class 6: Precision=0.3072, Recall=0.7837, F1=0.4414, IoU=0.2832
Class 7: Precision=0.7342, Recall=0.4257, F1=0.5390, IoU=0.3689
Class 8: Precision=0.9769, Recall=0.9988, F1=0.9877, IoU=0.9757
```

After

```
...  Best model: epoch 30,  val mIoU = 0.6012

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Conclusion & Future Work

Conclusion:

Demonstrated that synthetic data can effectively train segmentation models, Model can understand terrain features for off-road autonomy

Future Scope

- Apply model to real-world autonomous vehicles (UGVs)
- Improve accuracy using advanced deep learning architectures
- Expand training with more diverse simulated environments